MIHIR MITHANI

ASIC / VLSI – Digital Design, Verification & Physical Design

Rajkot, Gujarat, India | mihir.mithani29@gmail.com | +91 9904146370

LinkedIn | GitHub | Portfolio

SUMMARY

Final-year B.Tech ICT student with a strong foundation in RTL design (Verilog/SystemVerilog) and hands-on experience in the RTL-to-GDSII physical design flow. Built EDA automation tools and a chip routing optimizer that combines A* pathfinding with NVIDIA cuOpt. Currently interning in VLSI physical design; competition winner and IEEE Student Member.

EDUCATION

B.Tech, Information and Communication Technology – Marwadi University Jul 2023 – Mar 2027
Rajkot, India; 7th Semester (CGPA : 8.88 / 10)

- **Relevant coursework:** Computer Architecture, Computer Arithmetic, CMOS Transistors & Circuits, Digital Design, VLSI, Embedded Systems (Jetson Nano), Machine Learning, Deep Learning.

12th Science (PCB) – Indian Public School Jun 2021 – May 2023
Surendranagar, India (Grade : 57.61)

TECHNICAL SKILLS

- **Digital Design & Verification:** RTL design, Verilog, SystemVerilog, self-checking testbenches, Computer Architecture, CMOS Circuits (NPTEL – Digital Design Using Verilog).
- **Physical Design (exposure):** RTL-to-GDSII flow, OpenROAD / ORFS, Floorplanning, Placement, Clock Tree Synthesis (CTS), Routing, Static Timing Analysis (concepts).
- **Languages & EDA Tools:** Python, TCL, Shell/Bash, C/C++; OpenROAD, KLayout, EasyEDA, Vim/GVim, Linux CLI.
- **GPU / HPC:** CUDA Python (Numba), CUDA C++, NVIDIA cuOpt.

EXPERIENCE

VLSI Physical Design Intern – Microcircuits Innovations Pvt. Ltd. Jun 2026 – Present

- Working through the ASIC physical design flow with OpenROAD in a Linux-based EDA environment.
- Automating EDA workflows with TCL and shell scripting.

Project Coordination & Phase Lead, Robotics Team – ICT, Marwadi University Sep 2025 – May 2026

- Led a 15-member team delivering robotics projects (SO-100 Arm, TurtleBot) using ROS, NVIDIA Isaac Lab and Isaac Sim; reduced operational costs by 30–40% through optimized ROS-based communication pipelines.

Hardware Intern – eInfochips (An Arrow Company) Jun 2025 – Aug 2025

- Designed and validated an IoT-based hardware prototype for restaurant operations, integrating sensor modules and relay-based control logic.
- Analyzed existing operational and debugging workflows and implemented improvements that increased system performance by over 30%.

PROJECTS

OpenROAD Flow GUI – RTL-to-GDSII Desktop Automation Tool May 2026 – Present

- Built a Python/Tkinter desktop app that automates OpenROAD Flow Scripts (ORFS), the open-source RTL-to-GDSII flow; a PDK-aware wizard auto-generates design directories (Verilog, SDC) for ASAP7, Sky130, Nangate45 and GF12.

Synchronous FIFO – RTL Design & Verification Aug 2026

- Designed a parameterized synchronous FIFO in SystemVerilog with full/empty flags and overflow/underflow protection.
- Built a self-checking testbench with a queue-based reference model (directed tests plus 500 randomized operations), scoreboarding 245 reads with 0 mismatches; validated the testbench through deliberate bug injection.

Chip Routing Optimizer using NVIDIA cuOpt May 2026 – Present

- Built a PCB-style chip routing optimizer (octilinear A* pathfinding) integrated with NVIDIA cuOpt's VRP API to minimize total wire length across the layout.

CERTIFICATIONS, HONORS & LEADERSHIP

Certifications: Digital Design Using Verilog – NPTEL | Fundamentals of Accelerated Computing with CUDA Python – NVIDIA DLI | Getting Started with AI on Jetson Nano – NVIDIA DLI

Honors: Rank 1, CodeJam Sem 6 (GeeksforGeeks) | Winner, Byte Hunt IoT Trail – Circuitology Club